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Semester: III	**Year : AY 2026-27**
Subject: Programming with Java	
ASSIGNMENT No : 14	
TITLE : Exception Handling using Try-Catch	

Problem Statement

Write a Java program to demonstrate exception handling using try and catch blocks, nested and multiple try-catch blocks.

Learning Objectives

To handle runtime exceptions using try-catch, nested try, and multiple catch blocks.

Programme

Exercise

1. **Create an ATM program handling invalid withdrawal amount using try-catch**

Code:

```
import java.util.Scanner;

public class Main {
   public static void main(String[] args) {
      Scanner sc = new Scanner(System.in);

      double balance = 10000;
      System.out.print("Enter withdrawal amount: ");

      try {
         double amount = sc.nextDouble();

         if (amount <= 0) {
            throw new Exception("Invalid withdrawal amount!");
         }

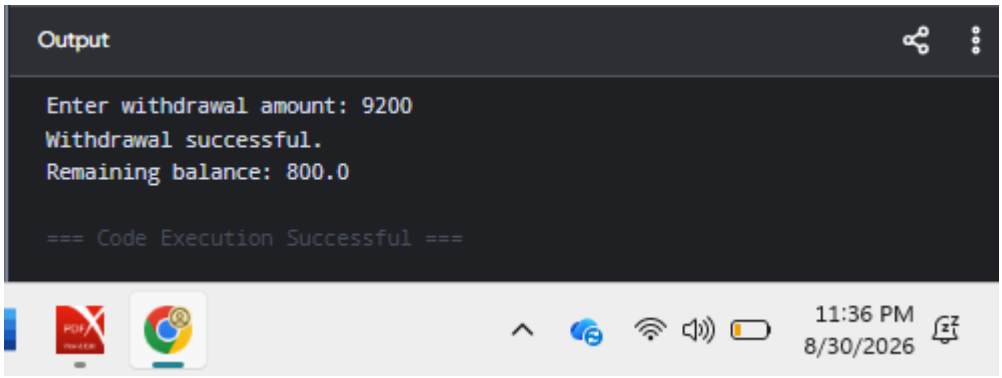
         if (amount > balance) {
            throw new Exception("Insufficient balance!");
         }

         balance = balance - amount;
```

```
            System.out.println("Withdrawal successful.");
            System.out.println("Remaining balance: " + balance);
        }
        catch (Exception e) {
            System.out.println("Error: " + e.getMessage());
        }

        sc.close();
    }
}
```

Output



2. **Create an Online Shopping program that handles an invalid product quantity entered by the user using a try-catch block. Display an appropriate error message if the quantity is less than or equal to zero.**

Code:

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter product name: ");
        String product = sc.nextLine();

        System.out.print("Enter product quantity: ");

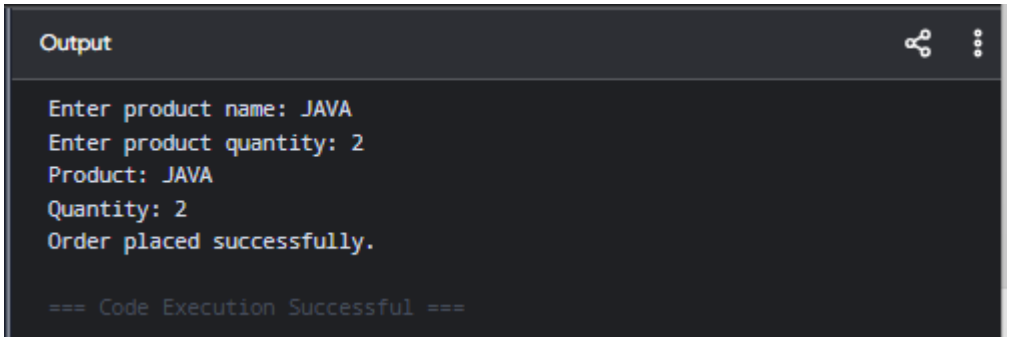
        try {
            int quantity = sc.nextInt();

            if (quantity <= 0) {
                throw new Exception("Invalid quantity! Quantity must be greater than zero.");
            }

            System.out.println("Product: " + product);
```

```
            System.out.println("Quantity: " + quantity);
            System.out.println("Order placed successfully.");
        }
        catch (Exception e) {
            System.out.println("Error: " + e.getMessage());
        }

        sc.close();
    }
}
```



Conclusion

The programs demonstrate the use of try-catch blocks to handle runtime exceptions effectively. The ATM program handles invalid withdrawal amounts and insufficient balance, while the Online Shopping program checks for invalid product quantities. If an invalid value is entered, an appropriate error message is displayed. Exception handling prevents the program from terminating unexpectedly and makes it more reliable and user-friendly. Thus, the use of try-catch provides an effective way to handle errors during program execution.